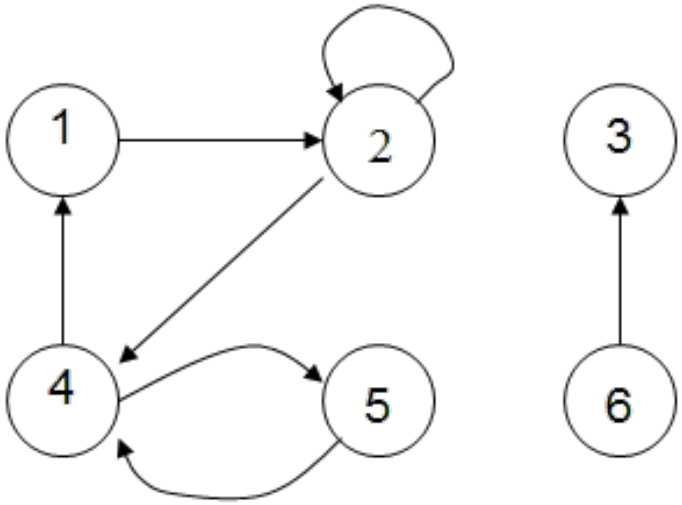
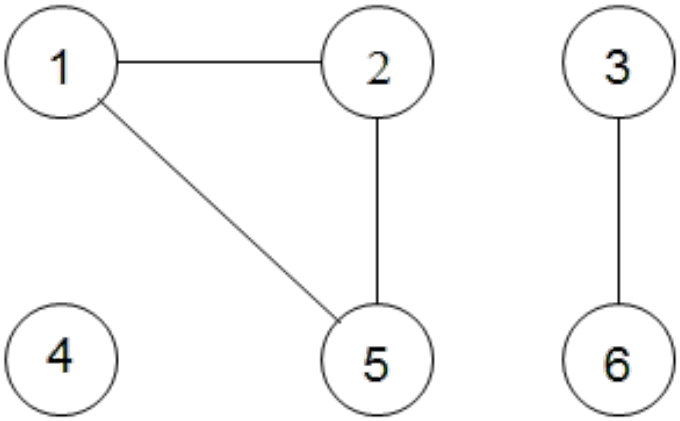
WQF7003
Intelligent Computation

Graph and Tree - A Revision

Graphs

- Graph → Network
- Informally a graph is a set of nodes joined by a set of lines or arrows.
- Representing a problem as a graph can make a problem much simpler.

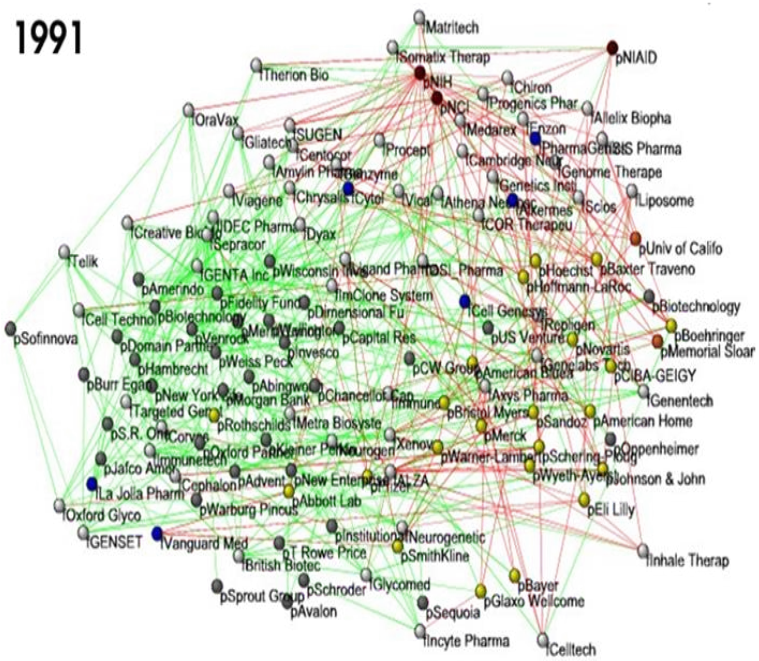




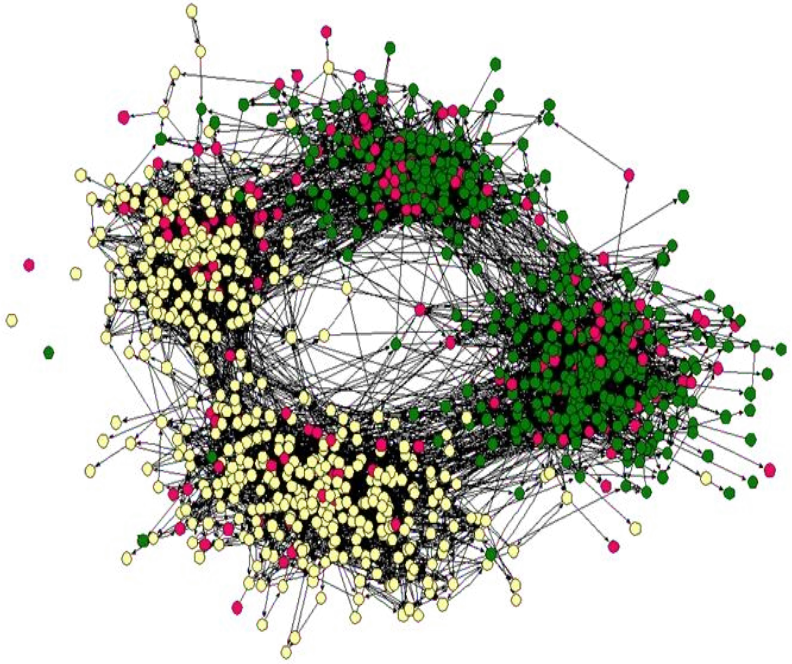
Application of Graphs

- In real world applications, graph is the base for all kinds of network.

Business ties in US biotech-industry

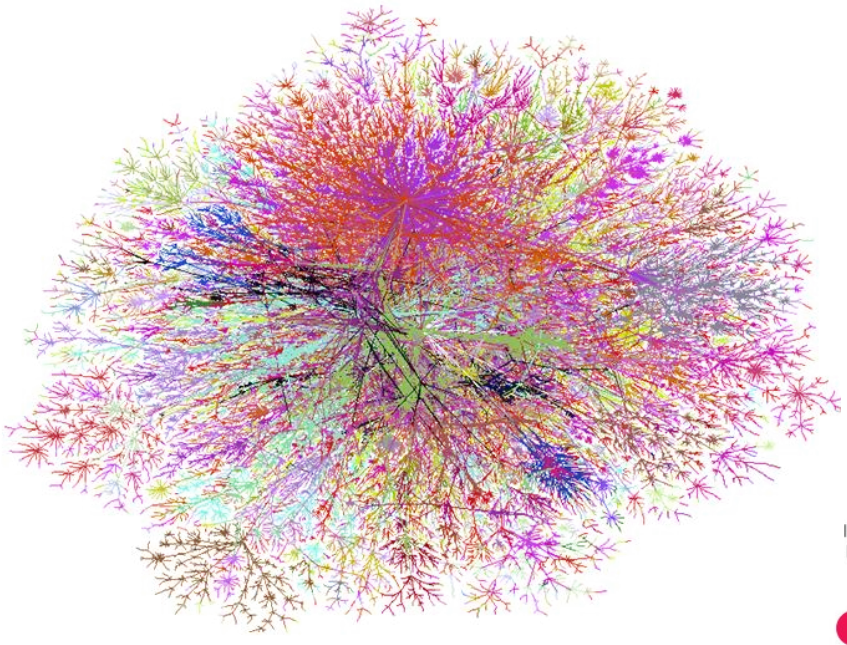


Friendship Network

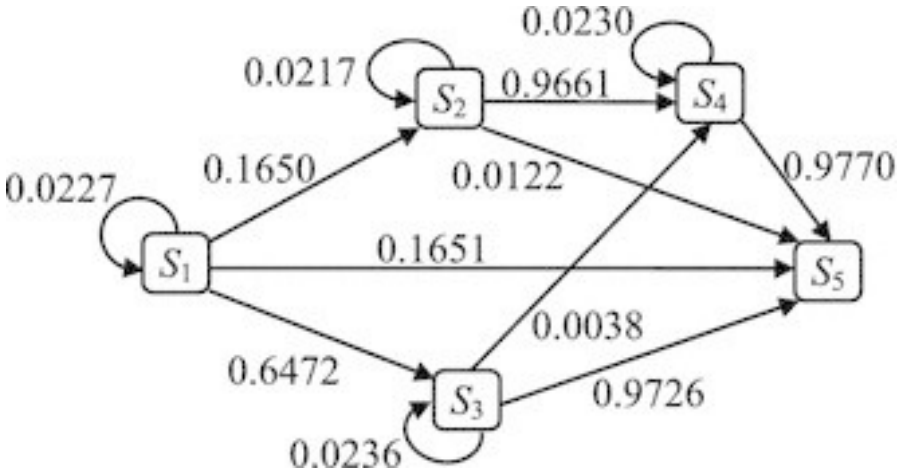


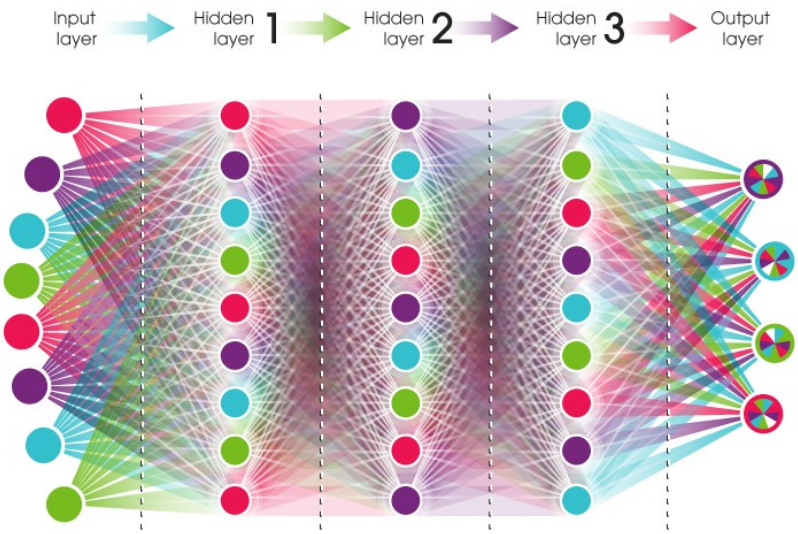
Application of Graphs in Computer Science

Internet



State Space Analysis





Neural Network

What makes a problem graph-like?

- There are two components to a graph
  - Vertex/Nodes
  - Edges

- In graph-like problems, these components have natural correspondences to problem elements
  - Entities are nodes and interactions between entities are edges
  - Many complex systems are graph-like

Definition

- A graph (specifically, undirected graph) G:
  - consists of a set V of vertices, and
  - a set E of edges
  - each edge $e \in E$ is associated with an **unordered** pair of vertices.

- A directed graph (or digraph) G :
  - consists of a set V of vertices, and
  - a set E of edges
  - each edge $e \in E$ is associated with an **ordered** pair of vertices.

Vertex and Edge

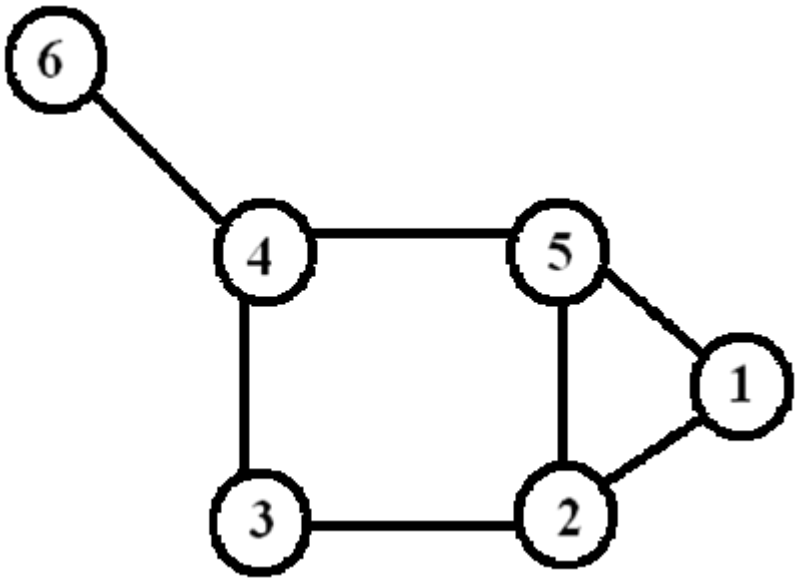
- **Vertex**
  - Drawn as a node or a dot.
  - Vertex set of G is usually denoted by V(G), or V
- **Edge**
  - Drawn as a line connecting two vertices
  - Represent relation between vertices
  - The edge set of G is usually denoted by E(G), or E.
  - If edge *e* is associated with vertices *p* and *q*:

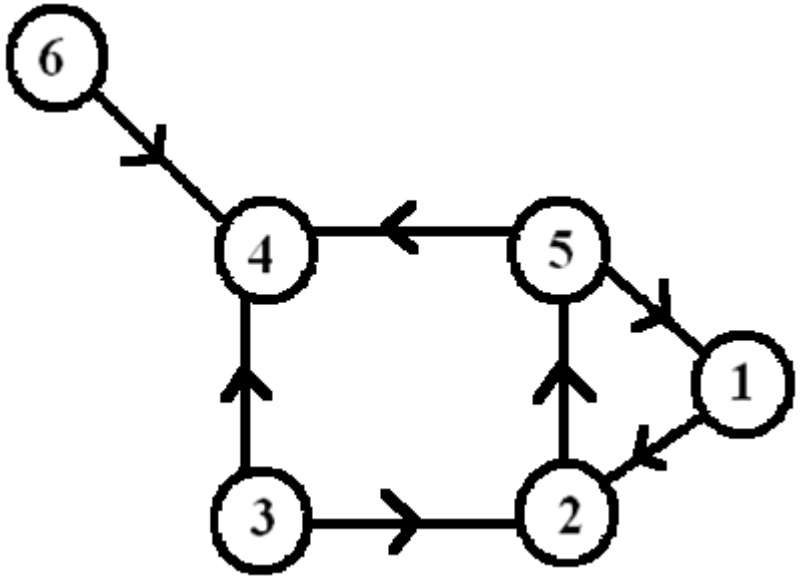
    "Edge *e* is **incident** on *p* and *q*"

    "*p* and *q* are **adjacent** vertices."

- Graph G with vertices V and edges E, **G=(V,E)**

Examples





Undirected Graph

V = { 1, 2 ,3, 4, 5, 6}

E = { {1,2}, {1,5}, {2,5}, {2,3}, {3,4}, {5,4}, {6,4} }

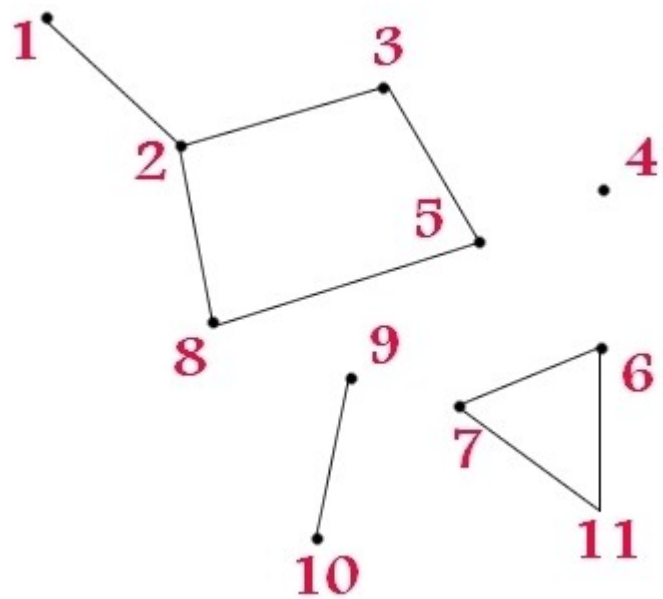
Directed Graph

V = { 1, 2 ,3, 4, 5, 6}

E = { (1,2), (5,1), (2,5), (3,2), (3,4), (5,4), (6,4) }

Isolated Vertices

- An isolated vertex is a vertex that is with not incident with any edge.
- Example : “4” is an isolated vertex.



Paths

- If, from vertex *p*, after travel along 1 or more edges, we eventually reach vertex *q:*

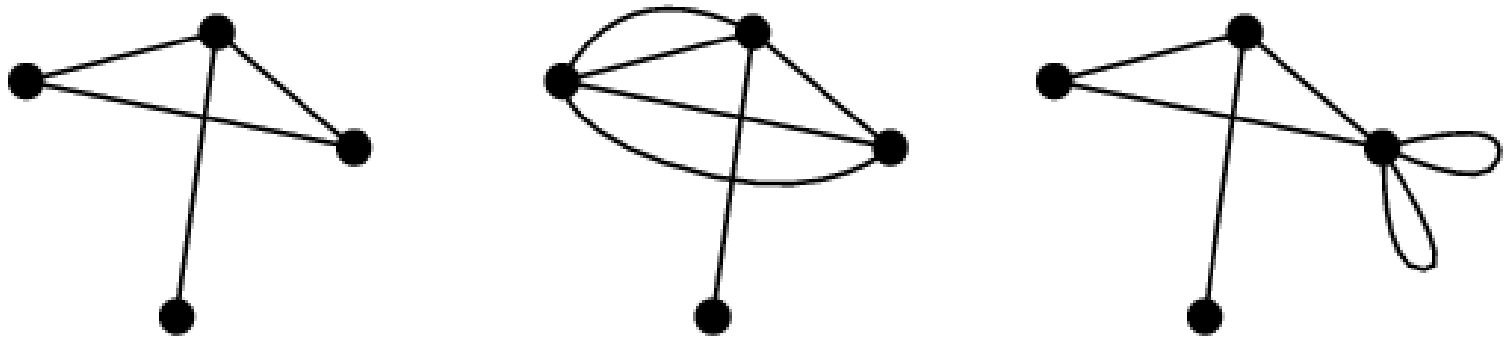
  "there is a **path** from *p* to *q*"

- Example (refer page 8):
  - On undirected graph, there is a path from any vertex to any vertex.
  - On directed graph, there is a path from (1) to (4)
  - On directed graph, there is no path from (4) to (1)

Type of Graphs

Simple Graphs

- Parallel/Multiple edges : edges that are both incident with the same vertex pair.
- Loop : An edge incident on a single vertex.
- Simple Graphs : graphs with no parallel edges, no loops



simple graph *nonsimple graph with multiple edges* *nonsimple graph with loops*

Weighted Graphs

- Graphs that have a number assigned to each edge are called weighted graphs.
- The number represents the weight of the edge.

Example: the weight of :

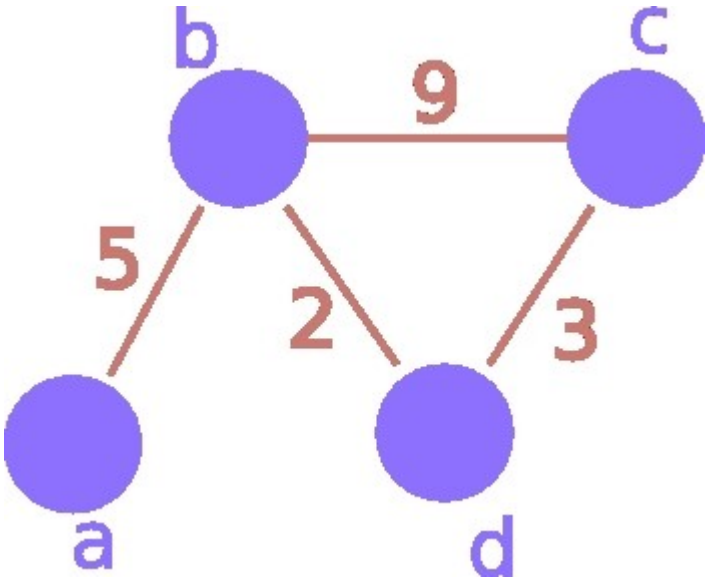
{a,b} = 5

{b,d} = 2

Length of path a to d

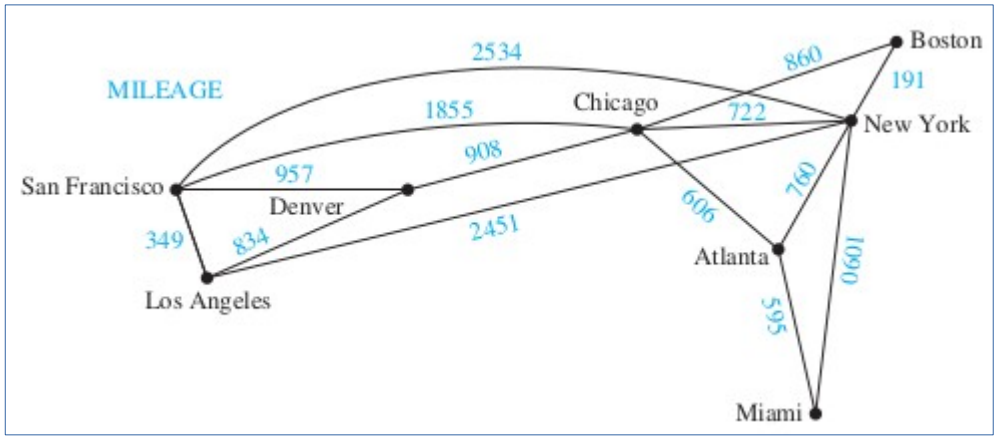
= 5+2 = 7

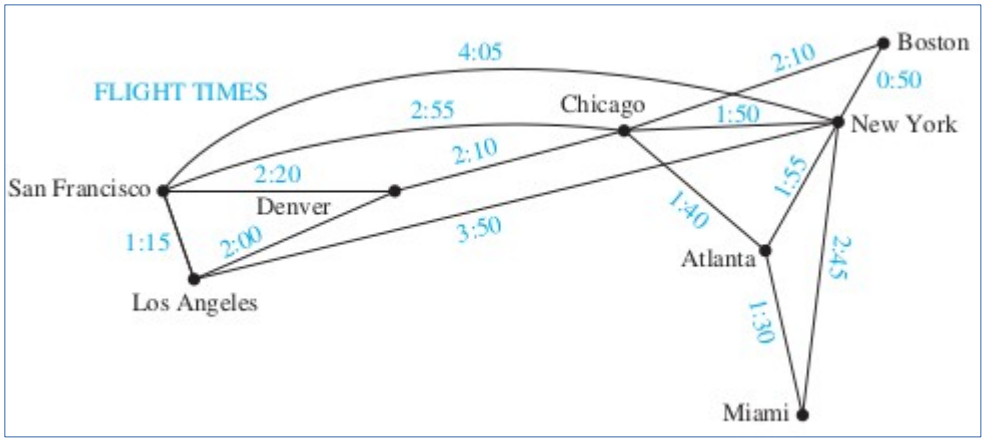
(If the weights represent length)

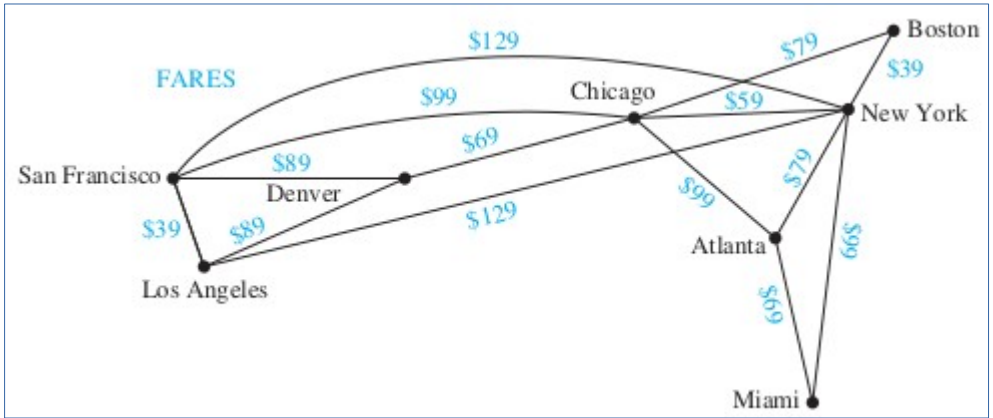


What is weight?

- Depends on what you want to model.
- Example: in an airline system, you get 3 different graphs if you assign different values to the graphs.





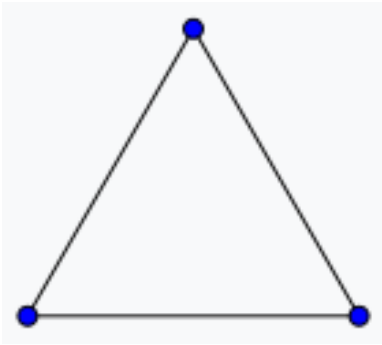


Complete Graphs

- A complete graph on n vertices, denoted K_n , is a **simple graph** with n vertices in which there is **an edge between every pair of distinct vertices**.
- Examples:

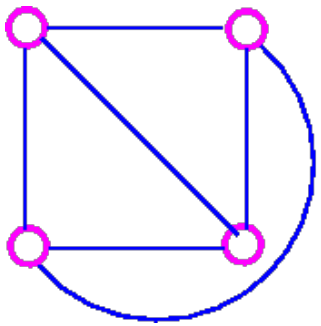
K_1

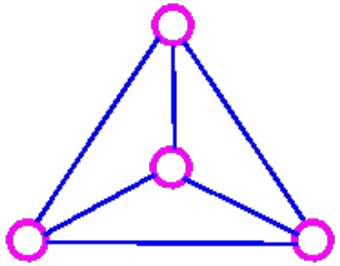
K_2



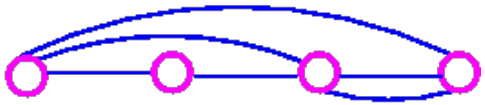
K_3

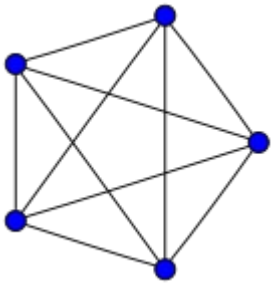
Complete Graphs



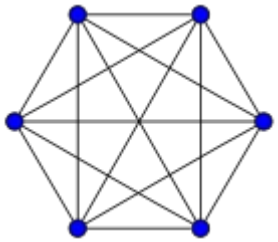


K_4





K_5



K_6

There are more ways to draw these complete graphs, but they represent the same since the “shape” of the graph doesn’t carry information in graph theory.

Exercise

How many edges can be found in K_5?

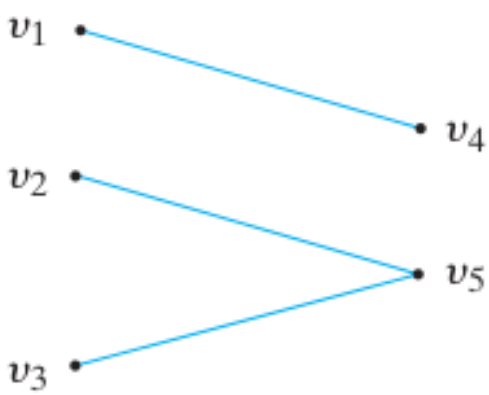
Bipartite Graphs

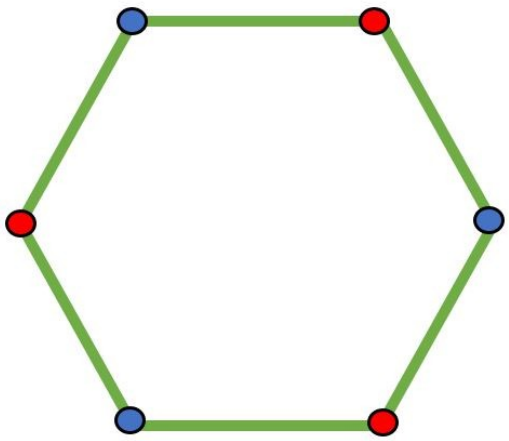
- Bipartite graph : a graph with all vertices decomposed into two disjoint sets V_1 and V_2, such that:

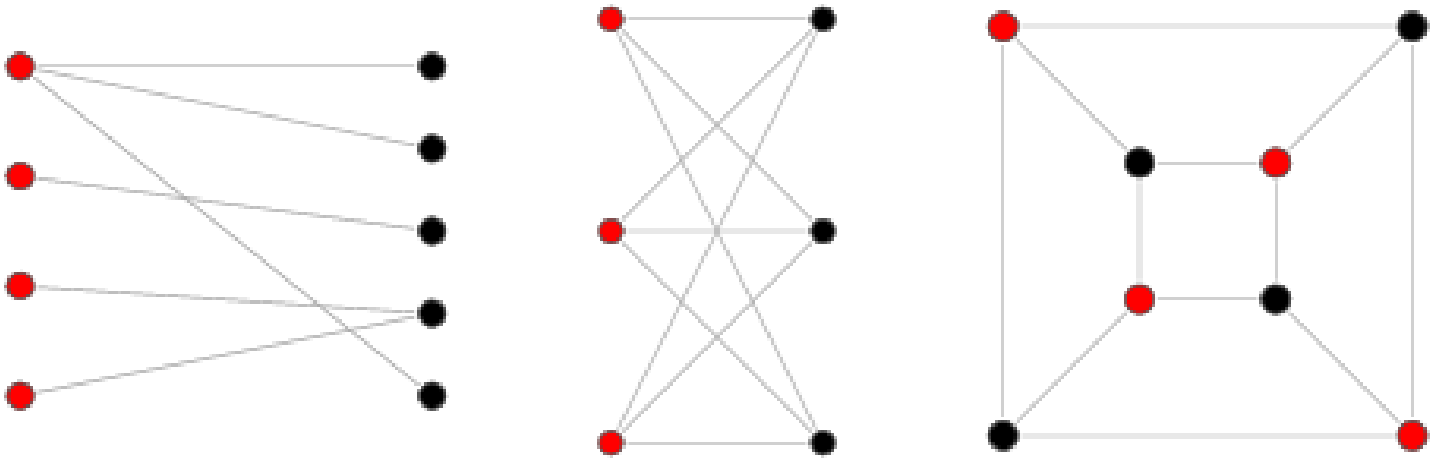
$$V_1 \cap V_2 = \emptyset \quad \text{and} \quad V_1 \cup V_2 = V$$

- Each edge on a bipartite graph incident one vertex in V_1 and one vertex in V_2.

Bipartite graphs (examples)

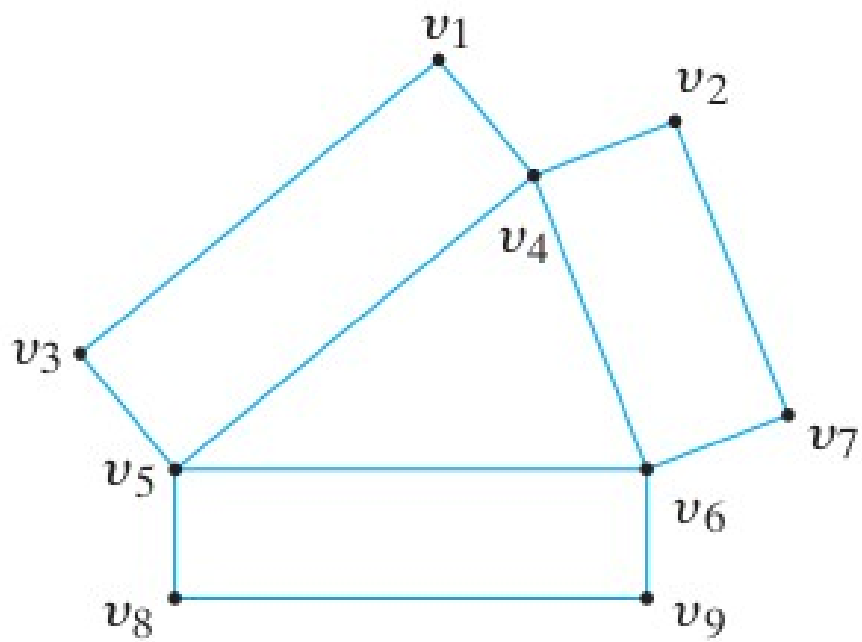






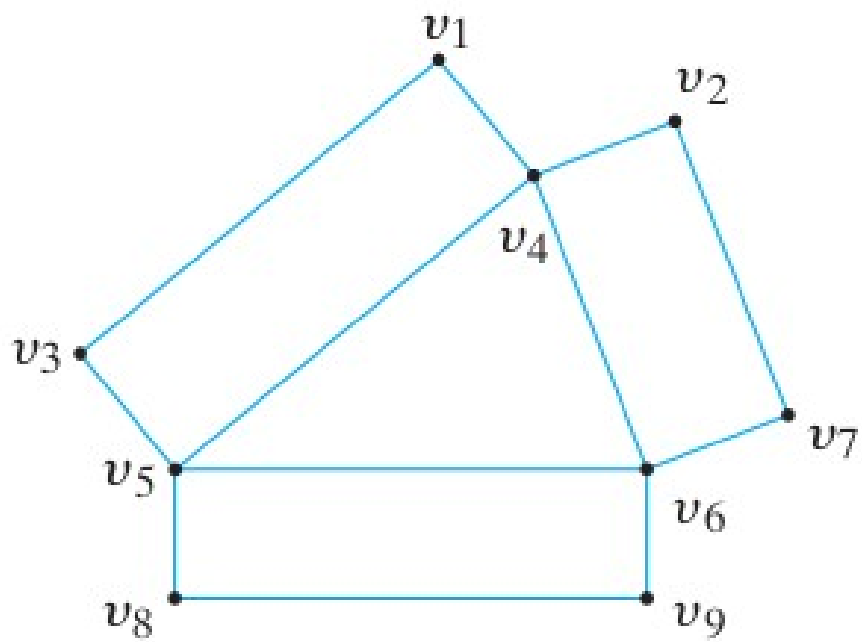
Bipartite graphs (exercise)

- Explain why the following graphs is not a bipartite graph:



Bipartite graphs (exercise)

- Explain why the following graphs is not a bipartite graph:



Clue : check v_4, v_5, v_6.

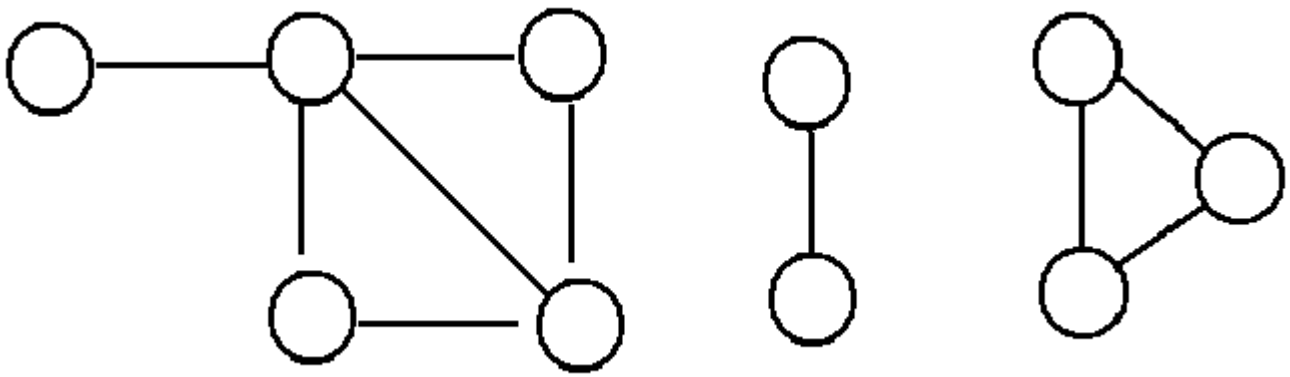
Components and Subgraphs

Connectivity

- A graph is connected if
  - you can get from any vertex to any other by following a sequence of edges OR
  - any two nodes are connected by a path.
- A directed graph is strongly connected if there is a directed path from any node to any other node.

Components

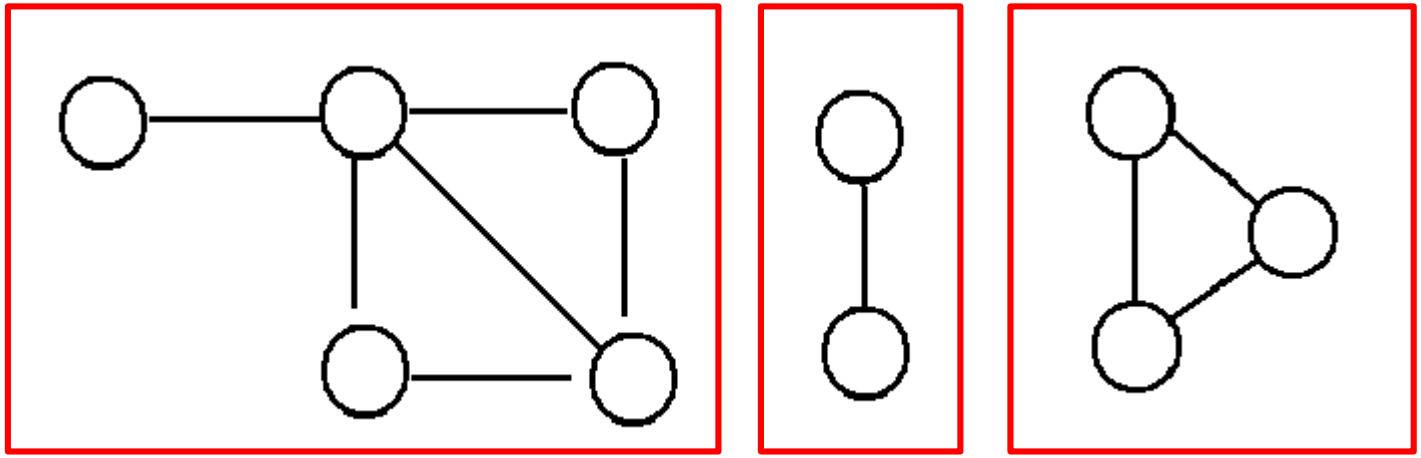
- Every disconnected graph can be split up into a number of connected components.



A graph with 10 vertices and 10 edges

Components

- Every disconnected graph can be split up into a number of connected components.

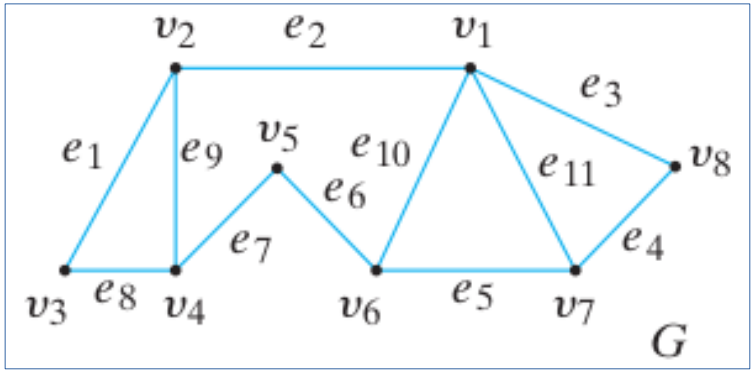


A graph with 10 vertices and 10 edges

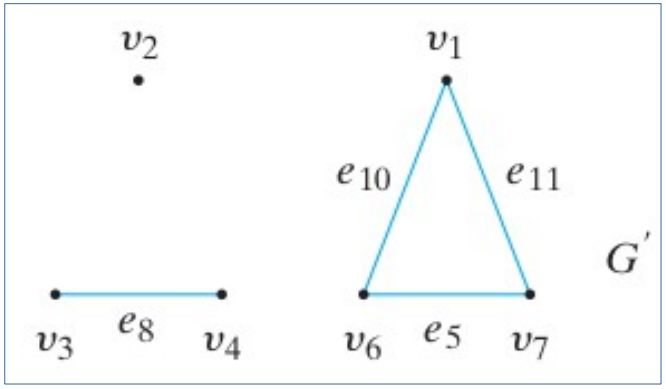
A graph with 3 components

Subgraphs

- A subgraph of a graph G = (V , E) is a graph H = (W, F), where W ⊆V and F ⊆E.
- A subset of vertices and edges of the original graph.
- Example: G' is a subgraph of G, with some edges and vertex removed.



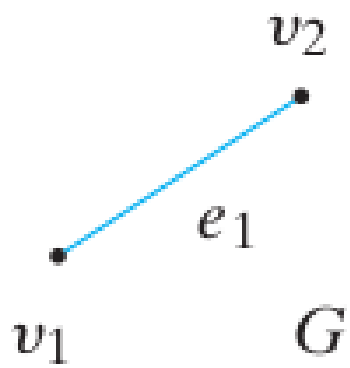
Graph G



One of the subgraph of G

Exercise

- Find all subgraphs of the graph G having at least one vertex.

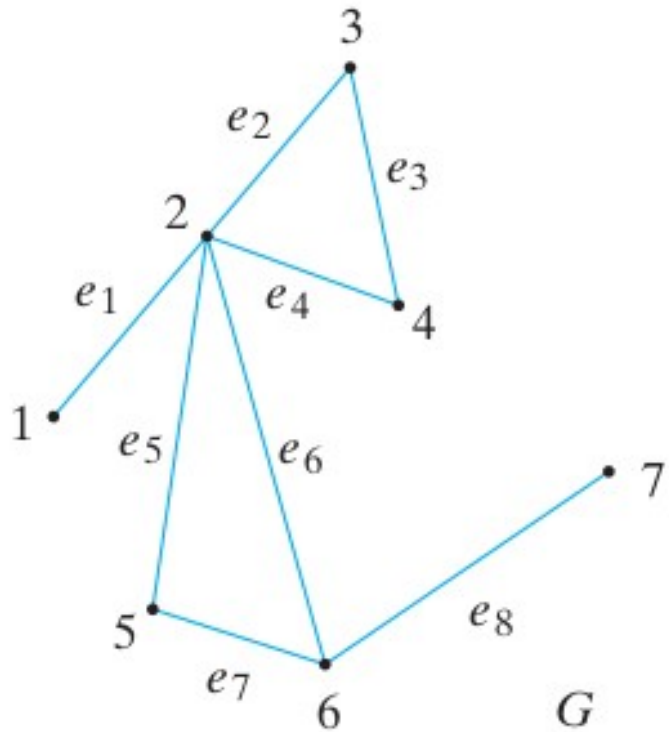


Paths and Cycles

Paths vs Cycles

- A **simple path** from *v* to *w* is a path from *v* to *w* with no repeated vertices.
- A **cycle** is a path of nonzero length from *v* to *v* with no repeated edges.
- A **simple cycle** is a cycle from *v* to *v* in which, except for the beginning and ending vertices that are both equal to *v*, there are no repeated vertices.

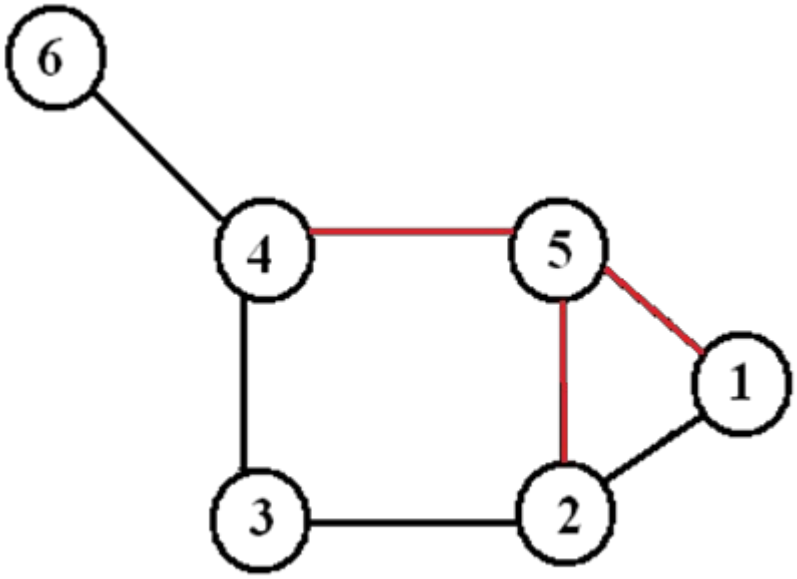
Example



Path	*Simple Path?*	*Cycle?*	*Simple Cycle?*
(6, 5, 2, 4, 3, 2, 1)	No	No	No
(6, 5, 2, 4)	Yes	No	No
(2, 6, 5, 2, 4, 3, 2)	No	Yes	No
(5, 6, 2, 5)	No	Yes	Yes
(7)	Yes	No	No

Degree

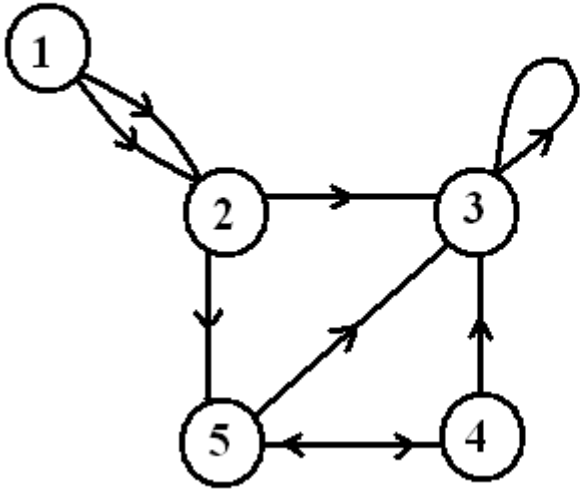
- Number of edges incident on a vertex.
- For simple graphs:



The degree of 5 is 3

Degree

- Number of edges incident on a vertex.
- For digraph:
  - In-degree: Number of edges entering
  - Out-degree: Number of edges leaving
  - Degree = indeg + outdeg



outdeg(1)=2
indeg(1)=0

outdeg(2)=2
indeg(2)=2

outdeg(3)=1
indeg(3)=4

Degree

- If G is a digraph then

$$\sum \text{indeg}(v) = \sum \text{outdeg}(v) = |E|$$

- If G is a graph with m edges, then

$$\sum \deg(v) = 2m = 2\,|E|$$

- Number of Odd degree Nodes is even.

Proof: $$\sum_{v \in V_{even}} \deg(v) + \sum_{v \in V_{odd}} \deg(v) = \text{Even Number}$$

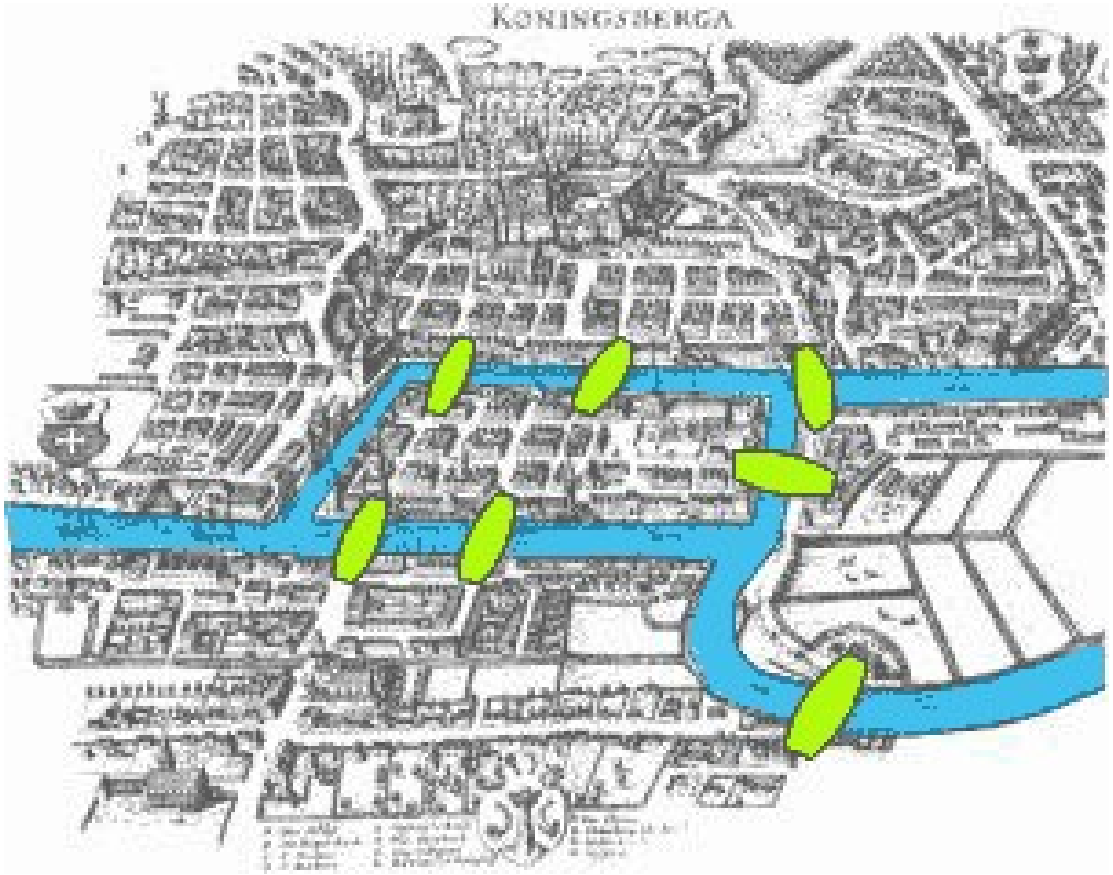
The first term is a sum of even numbers, so the result is even.

The second term is a sum of odd numbers, but the result is even, so the $|V_{odd}|$ is even as well.

Euler Cycles

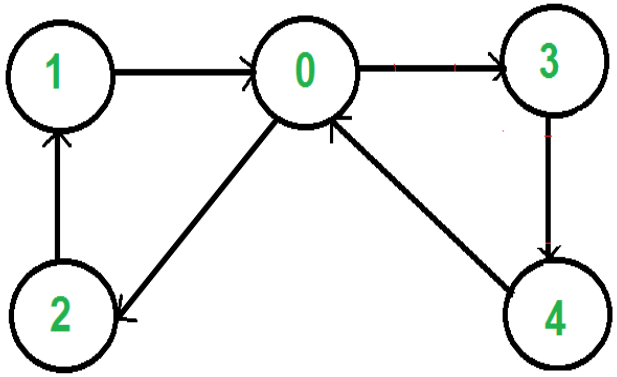
- Euler cycle - a cycle in a graph G that includes all of the edges and all of the vertices of G.
- A graph G has an Euler cycle if and only if all vertices of non-zero degree belong to a single connected component and every vertex has an even degree.
- If G has a Euler cycle and it has only a few edges, we can usually find the Euler cycle by inspection.

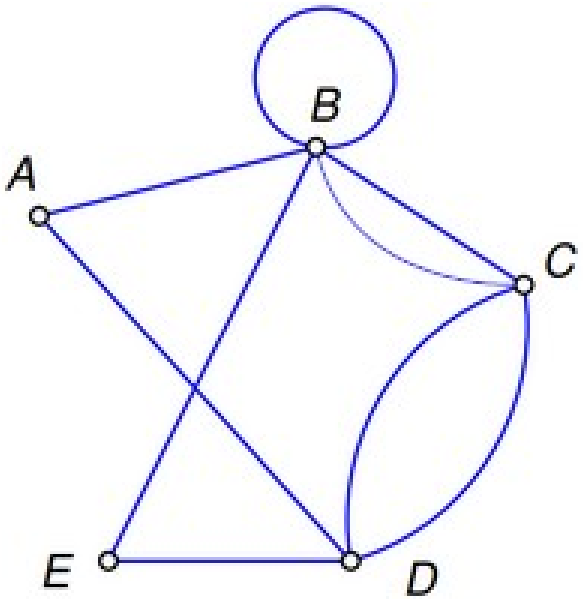
Euler and
Seven Bridges of Königsberg



Euler Cycles

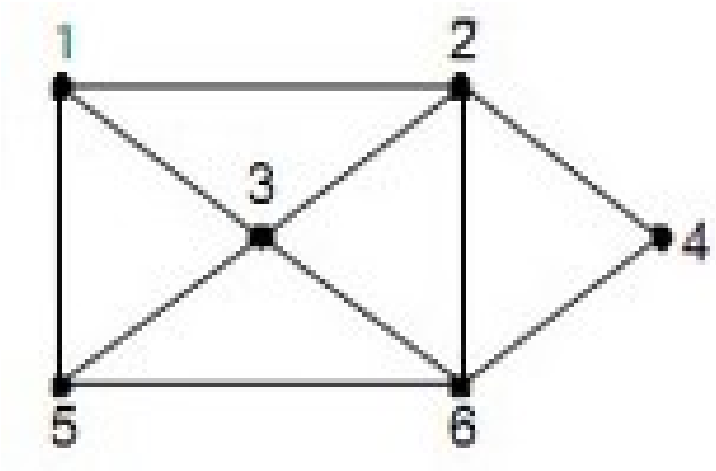
- Examples





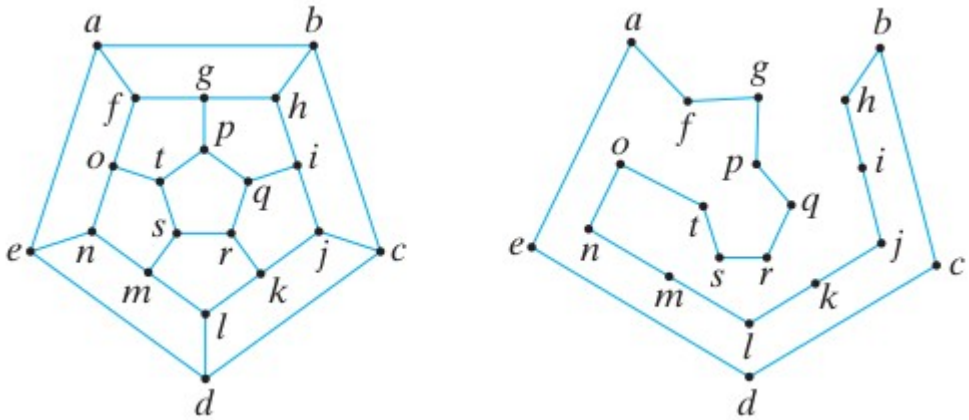
Additional info

- A graph has a path with no repeated edges from v to w (v ≠ w) containing all the edges and vertices if and only if it is connected and v and w are the only vertices having odd degree.
- Example:



Hamiltonian Cycles

- A cycle in a graph G that contains each vertex in G exactly once, except for the starting and ending vertex that appears twice.
- An Euler cycle visits each edge once, whereas a Hamiltonian cycle visits each vertex once.
- no easily verified necessary and sufficient conditions are known for the existence of a Hamiltonian cycle in a graph



A graph and its Hamiltonian cycle

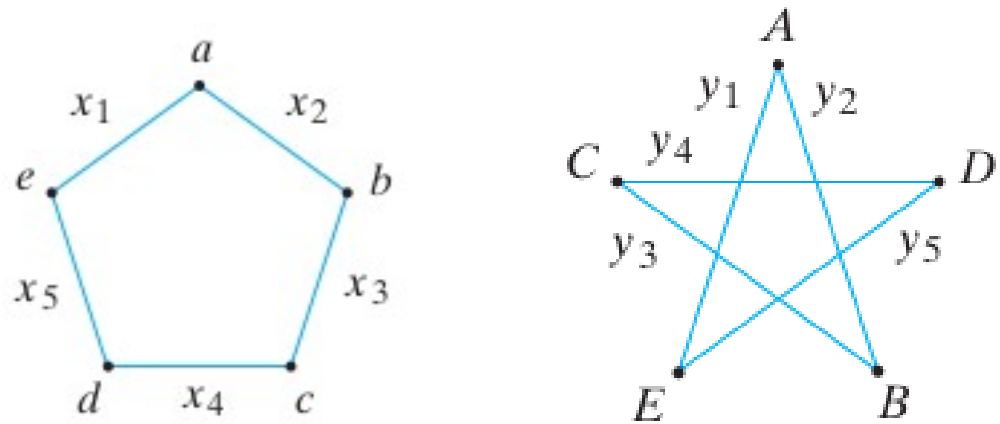
Traveling salesman problem

- Classical computer science problem.
- “Given a list of cities and the distances between each pair of cities, what is the shortest possible route that visits each city exactly once and returns to the origin city”
- Finding Hamiltonian cycles contribute solutions to this problem.

Representation of Graphs

Representation of Graphs

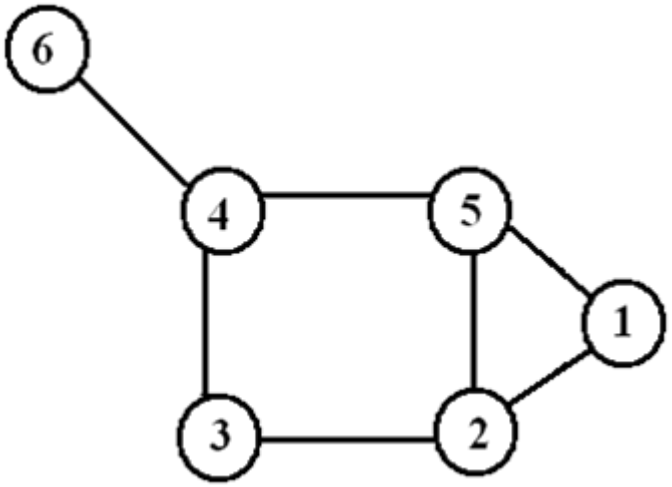
- Sometime, two graphs “look” different but actually they are the same.
- We say these graphs are isomorphic.
- Example:



- Despite the drawings are different, isomorphic graphs may find similar representation.

Representation with Matrix

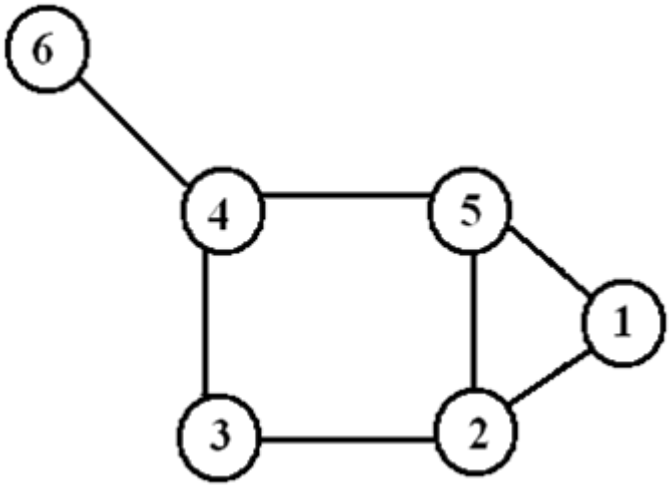
- Incidence Matrix
  - V x E
  - The rows represent the vertices and the columns represent the edges. The entry for row v and column e is 1 if e is incident on v and 0 otherwise.



$$
\begin{array}{c}
 \\ 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6
\end{array}
\begin{array}{c}
\begin{array}{ccccccc} 1,2 & 1,5 & 2,3 & 2,5 & 3,4 & 4,5 & 4,6 \end{array} \\
\begin{pmatrix}
1 & 1 & 0 & 0 & 0 & 0 & 0 \\
1 & 0 & 1 & 1 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 1 & 1 & 1 \\
0 & 1 & 0 & 1 & 0 & 1 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 1
\end{pmatrix}
\end{array}
$$

Representation with Matrix

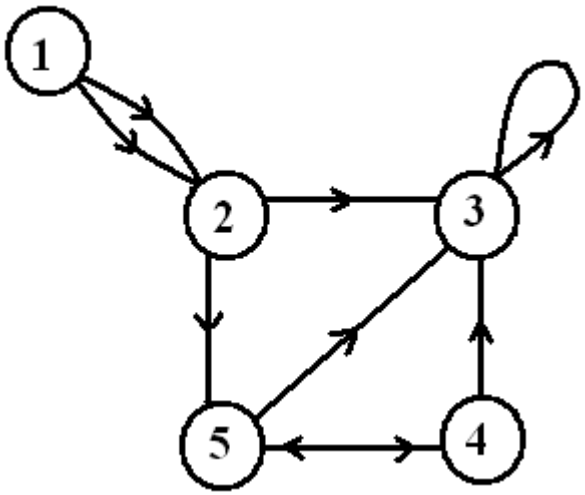
- Adjacent Matrix
  - V x V
  - Adjacent→1 not adjacent →0
  - Or Edge Weights



$$
\begin{array}{c|cccccc}
 & 1 & 2 & 3 & 4 & 5 & 6 \\
\hline
1 & 0 & 1 & 0 & 0 & 1 & 0 \\
2 & 1 & 0 & 1 & 0 & 1 & 0 \\
3 & 0 & 1 & 0 & 1 & 0 & 0 \\
4 & 0 & 0 & 1 & 0 & 1 & 1 \\
5 & 1 & 1 & 0 & 1 & 0 & 0 \\
6 & 0 & 0 & 0 & 1 & 0 & 0
\end{array}
$$

Representation with List

- Adjacency List (node list)
  - An array of |V| lists, one for each vertex in V.
  - For each v ∈ V, ADJ [v] points to all its adjacent vertices.

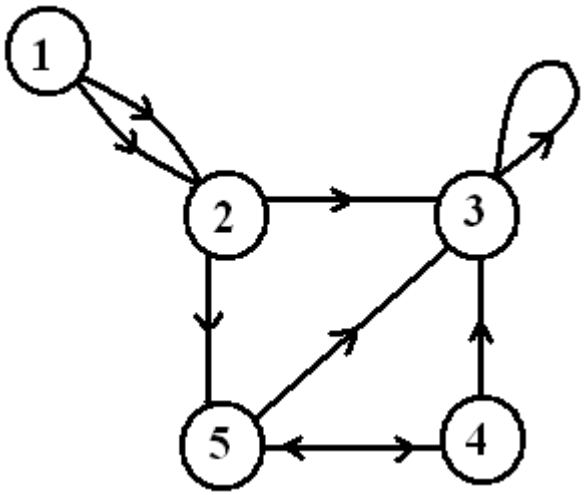


Node List

1 2 2
2 3 5
3 3
4 3 5
5 3 4

Representation with List

- Edge List
  - Pairs (ordered if directed) of vertices
  - Optionally weight and other data
  - Example 1:

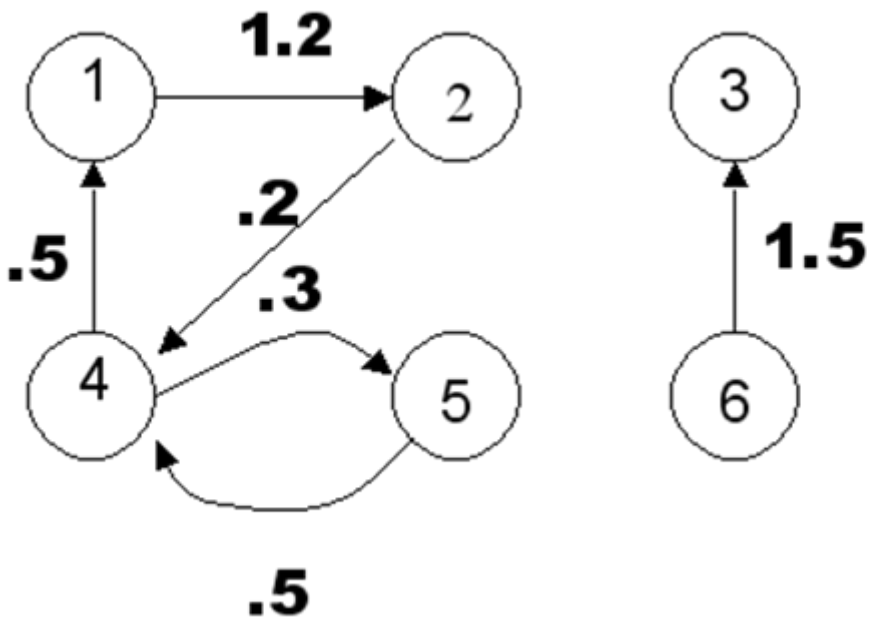


Edge List

1 2
1 2
2 3
2 5
3 3
4 3
4 5
5 3
5 4

Representation with List

- Edge List
  - Pairs (ordered if directed) of vertices
  - Optionally weight and other data
  - Example 2:



Edge List

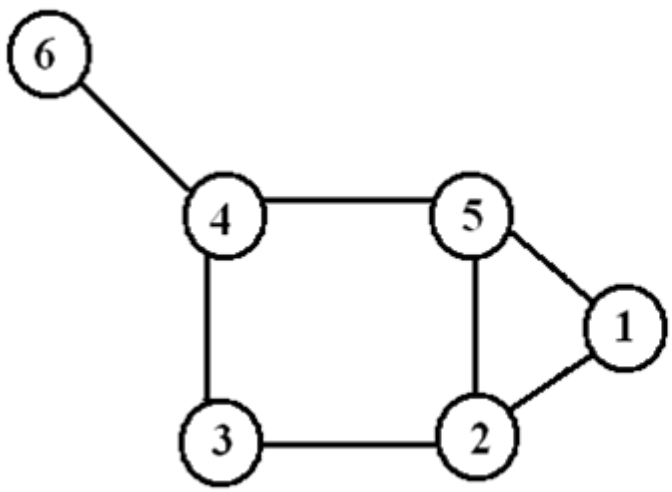
1 2 1.2
2 4 0.2
4 5 0.3
4 1 0.5
5 4 0.5
6 3 1.5

Topological Distance

- A shortest path is the minimum path connecting two nodes.
- The number of edges in the shortest path connecting *p* and *q* is the topological distance between these two nodes, $d_{p,q}$

Distance Matrix

- $|V| \times |V|$ matrix $D = (d_{ij})$ such that d_{ij} is the topological distance between i and j.

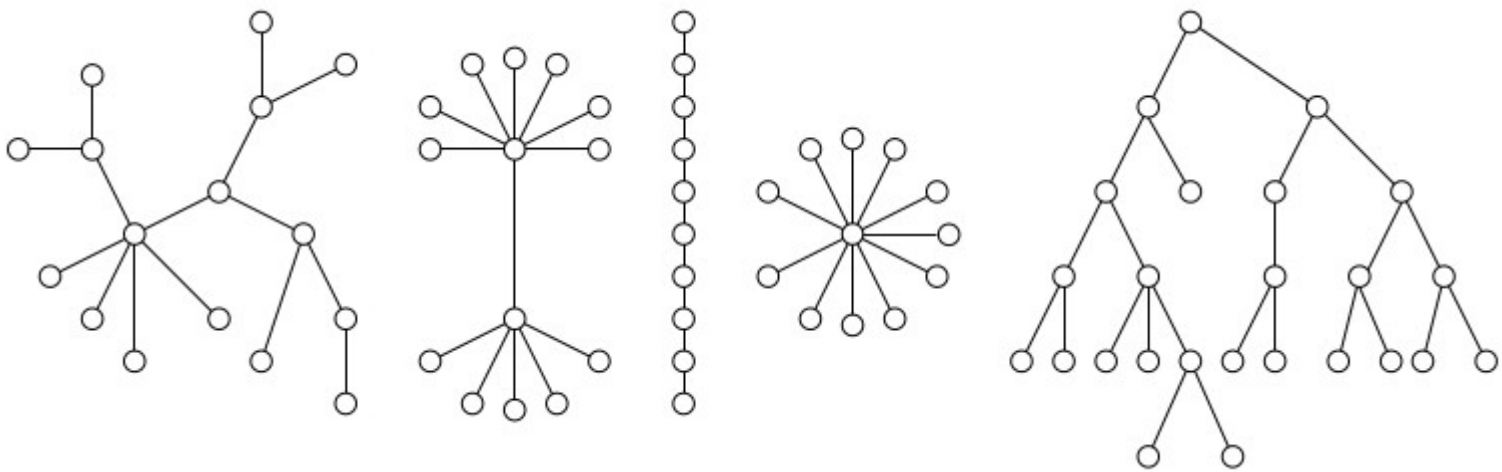


$$
\begin{array}{c|cccccc}
 & 1 & 2 & 3 & 4 & 5 & 6 \\
\hline
1 & 0 & 1 & 2 & 2 & 1 & 3 \\
2 & 1 & 0 & 1 & 2 & 1 & 3 \\
3 & 2 & 1 & 0 & 1 & 2 & 2 \\
4 & 2 & 2 & 1 & 0 & 1 & 1 \\
5 & 1 & 1 & 2 & 1 & 0 & 2 \\
6 & 3 & 3 & 2 & 1 & 2 & 0
\end{array}
$$

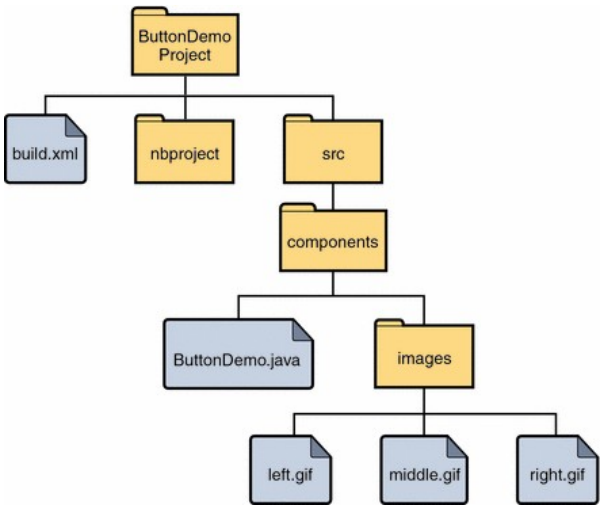
Trees

Tree

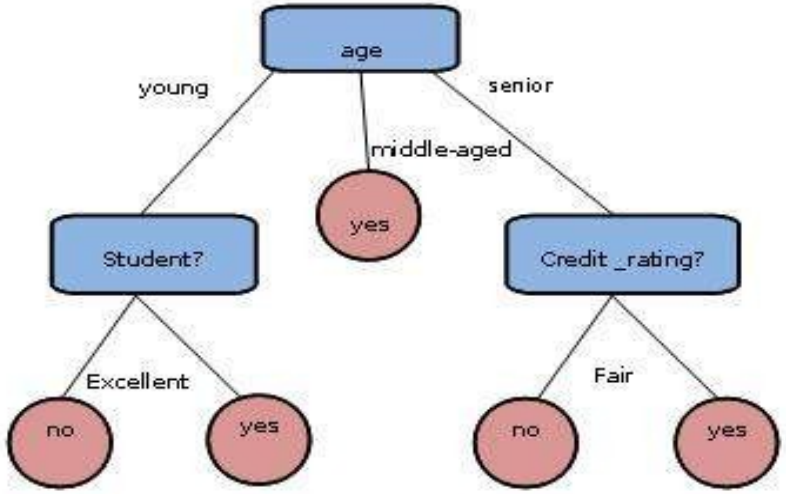
- A tree is a connected undirected graph with no simple circuits (cycles).
- cannot contain multiple edges or loops.
- Examples of tree:



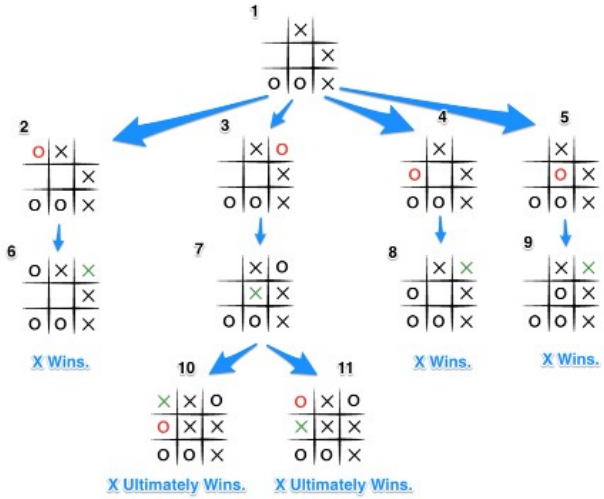
Where we find trees in Computer Science



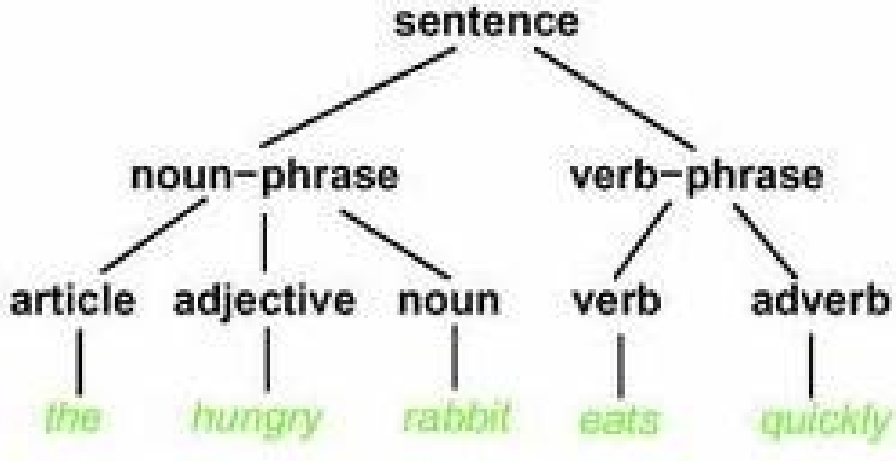
File structures



Decision Tree

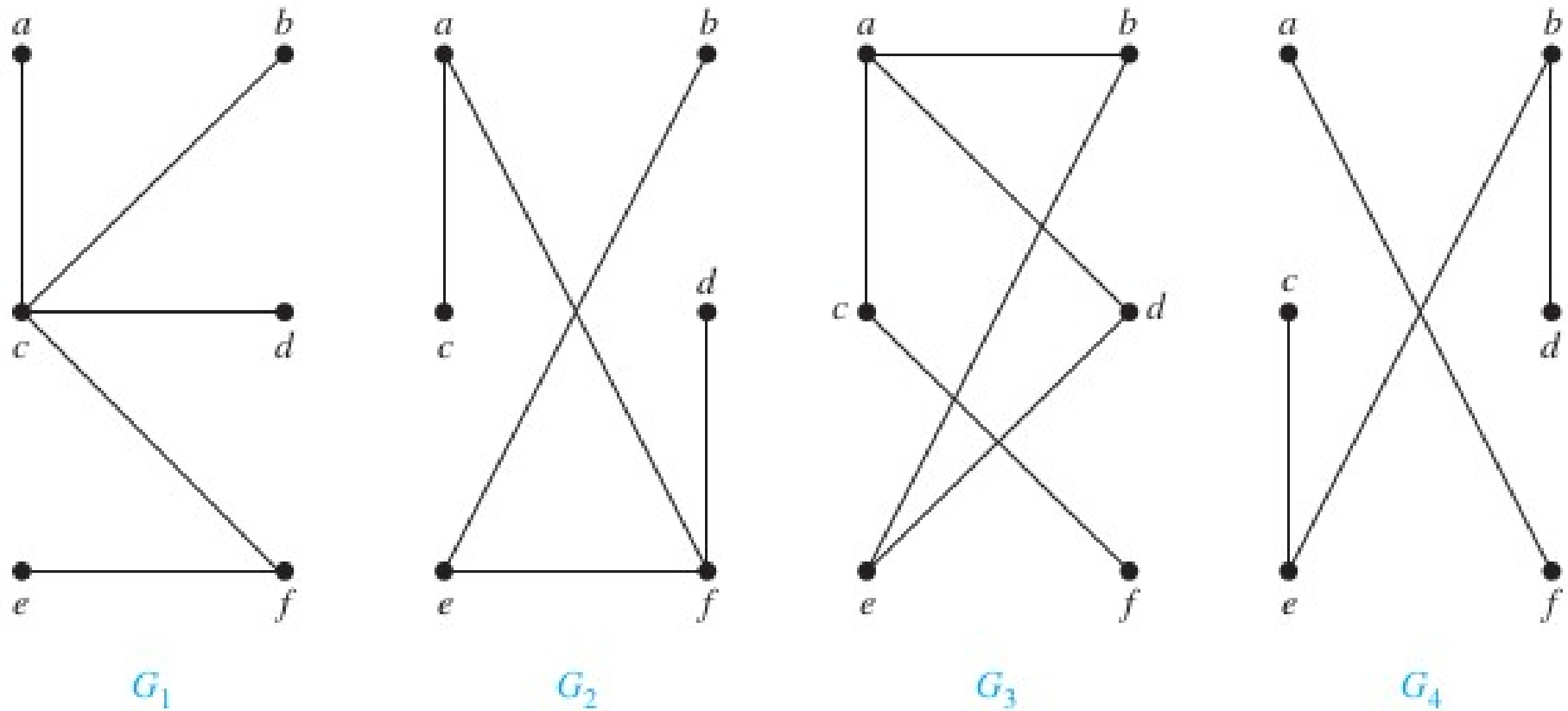


Game analysis

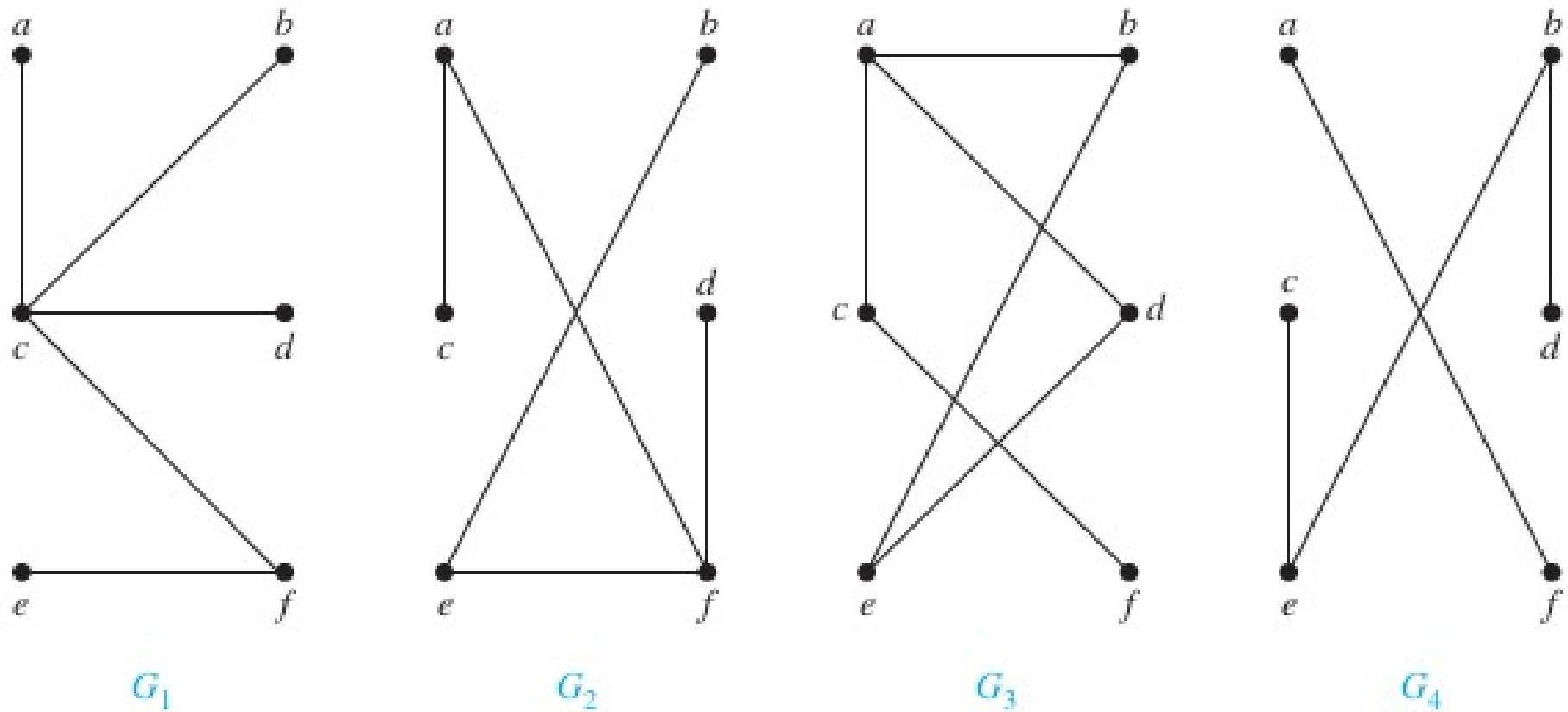


Parsing sentences

Exercise : Are these trees?



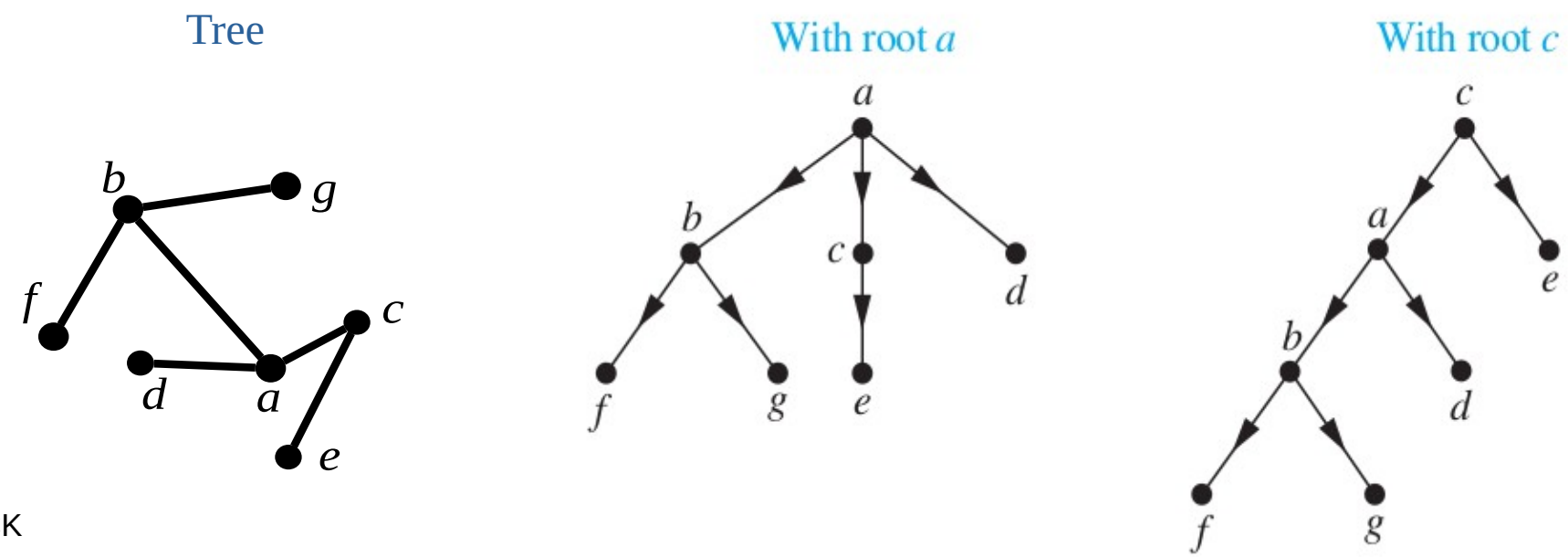
Exercise : Are these trees?



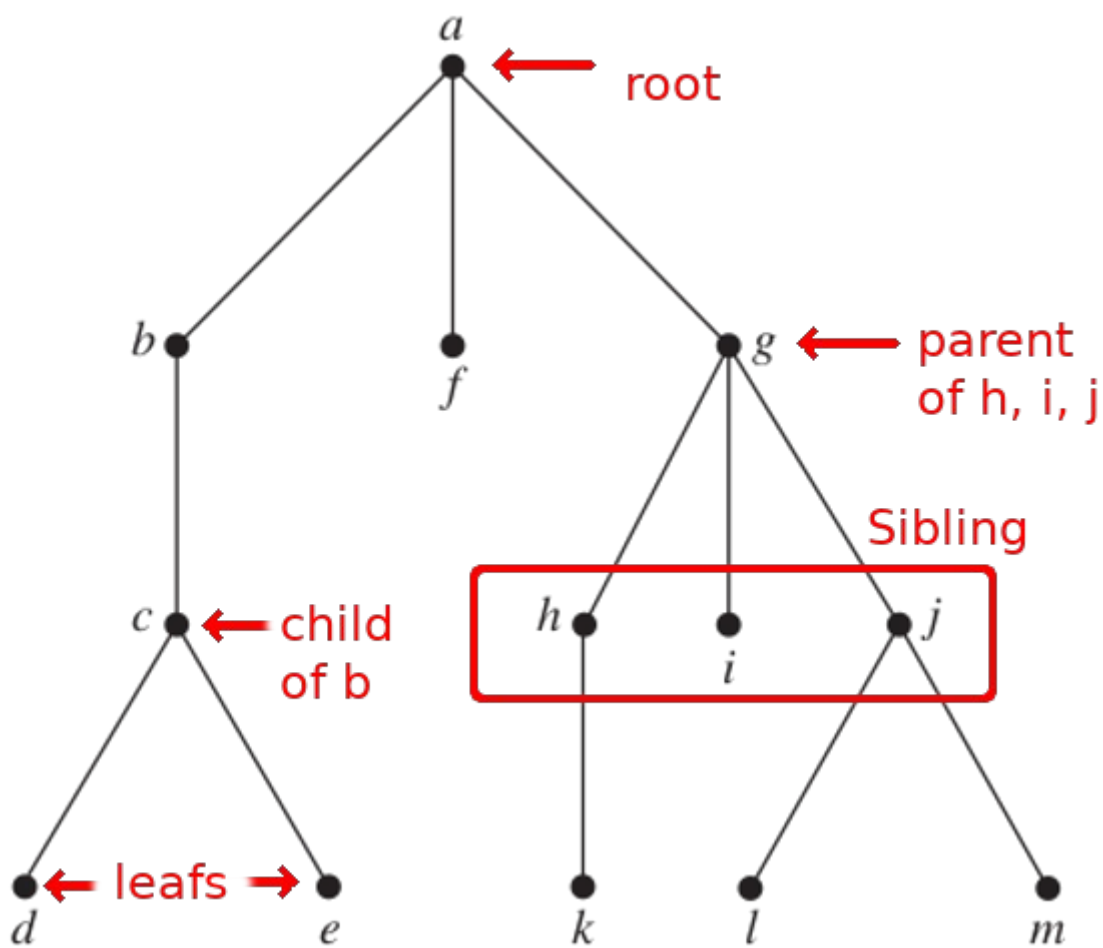
Tree | Tree | Not tree | Not tree

Rooted Trees

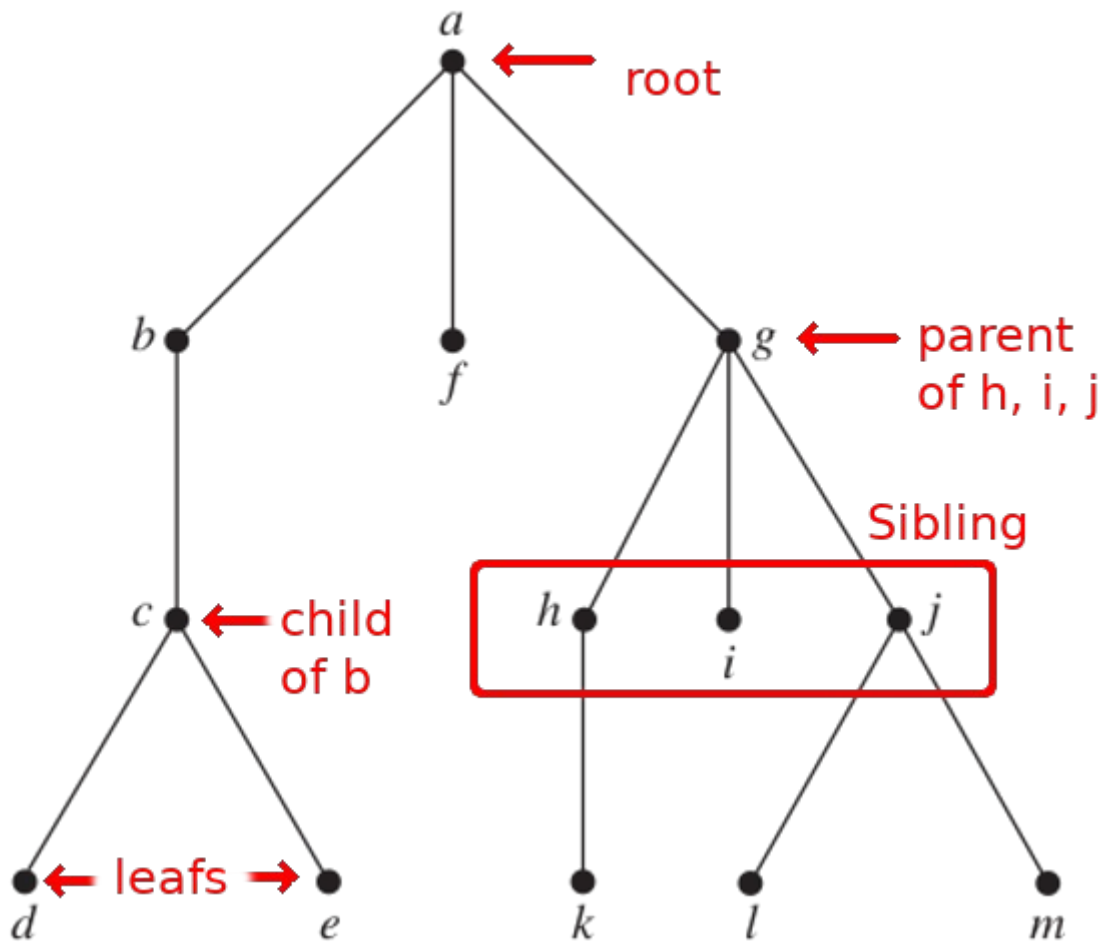
- A rooted tree is a tree in which one vertex has been designated as the root and every edge is directed away from the root.
- We usually draw a rooted tree with its root at the top of the graph.



Rooted trees - terminology

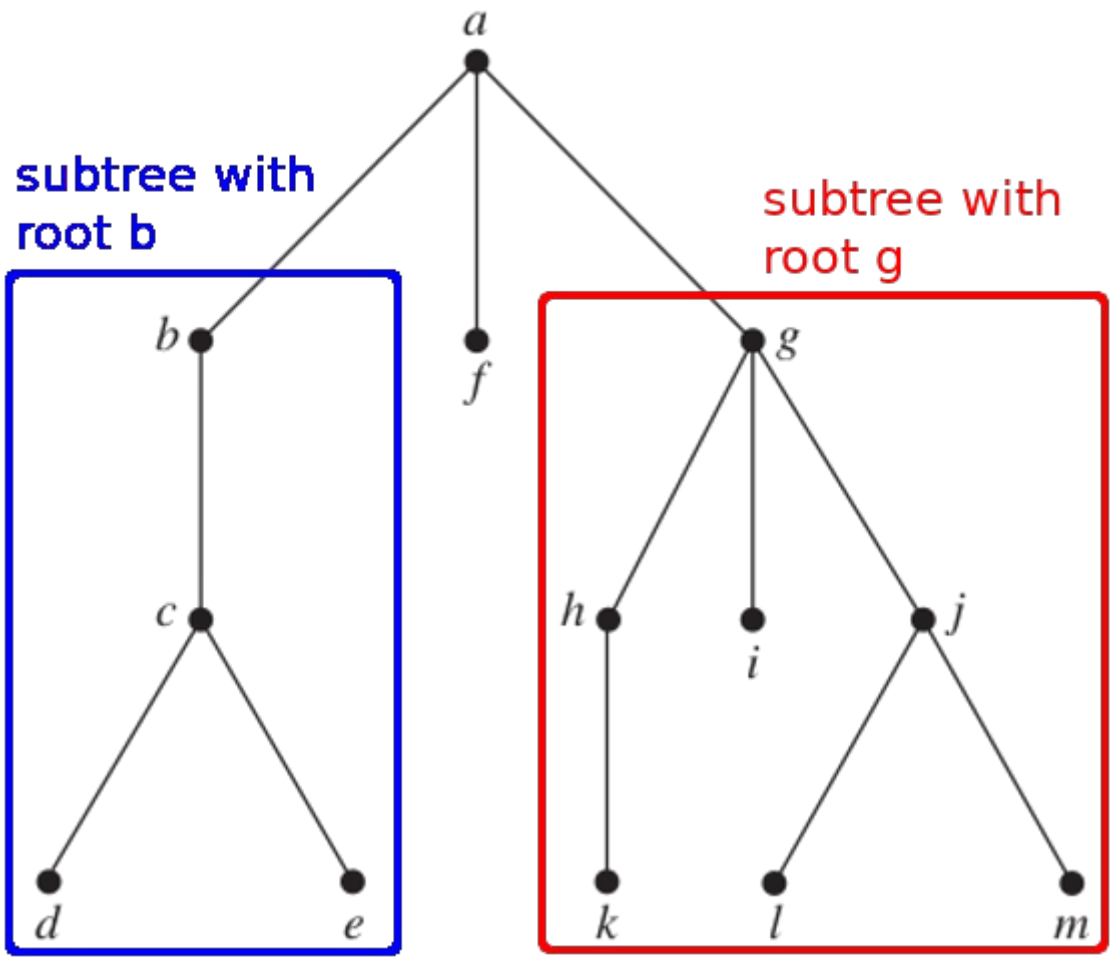


Rooted trees - terminology

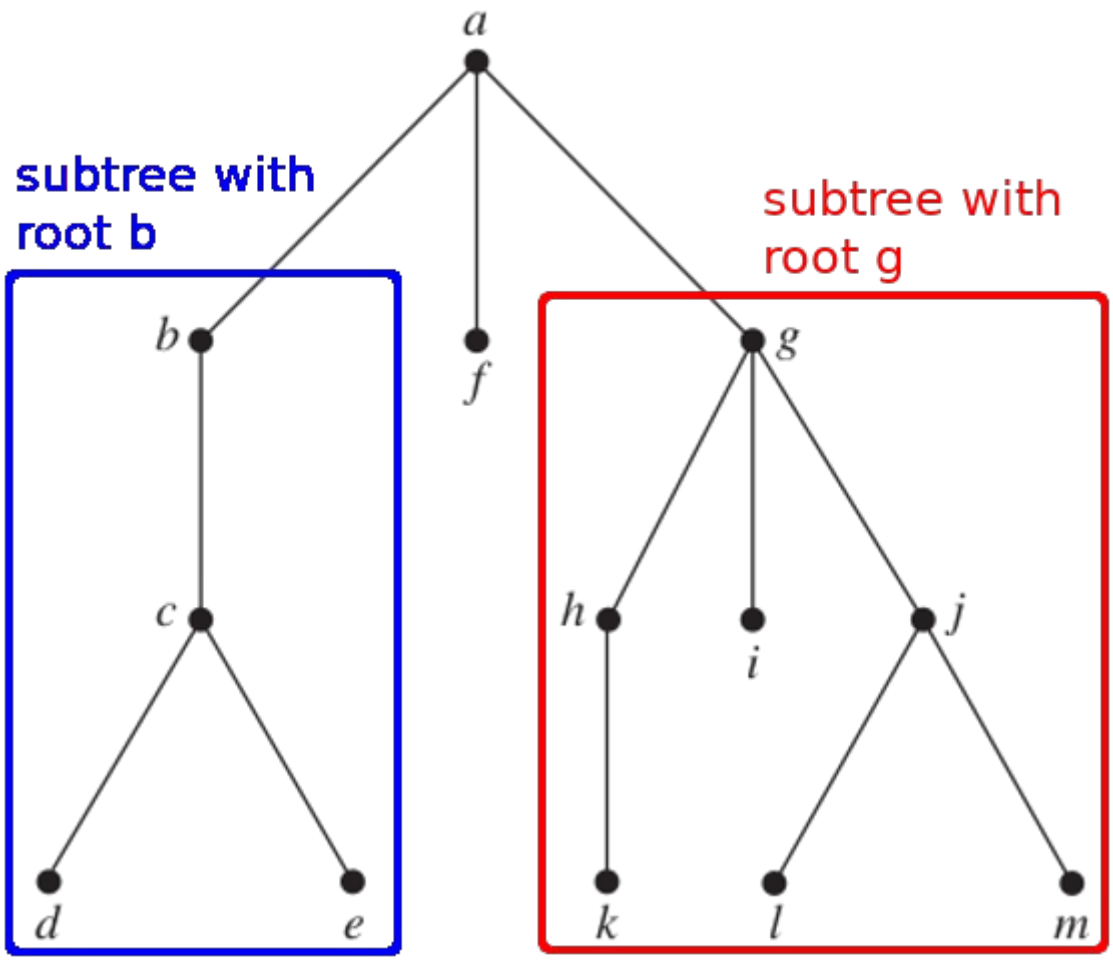


internal vertices (vertex with child) : a, b, c, g, h, j

Rooted trees - terminology

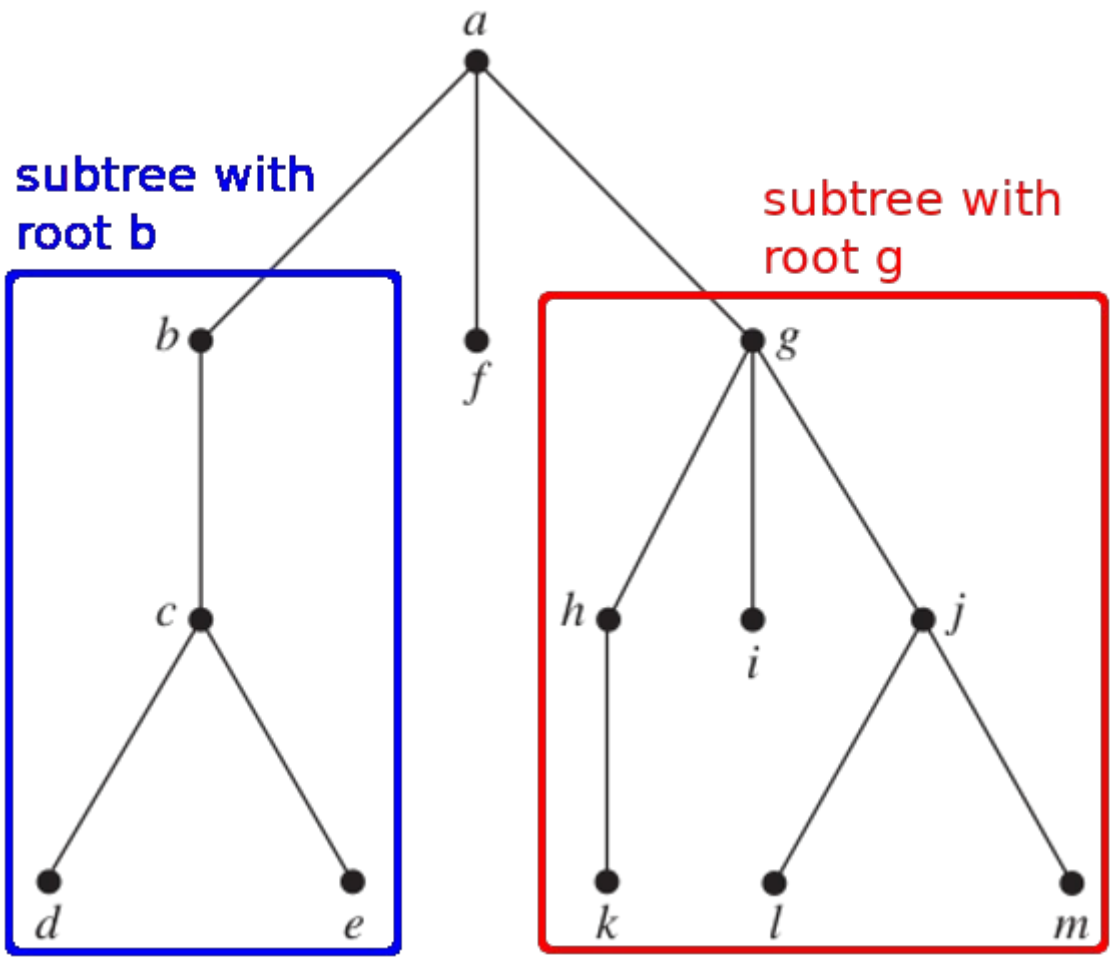


Rooted trees - terminology



Ancestors of e : c, b, a

Rooted trees - terminology

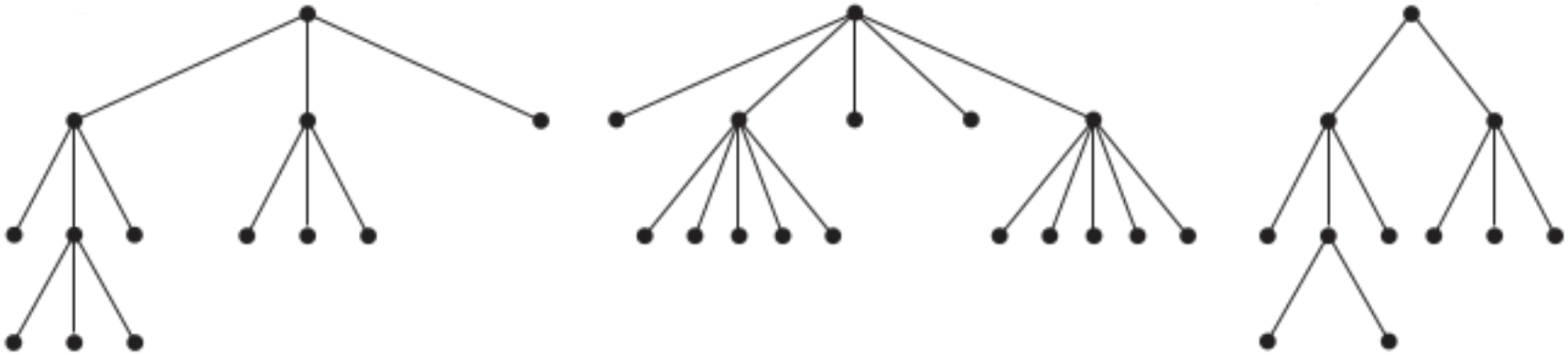


Ancestors of e : c, b, a

Descendants of g : h, i, j, k, l, m

m-ary trees

- A rooted tree is called an m-ary tree if every internal vertex has no more than m children.
- The tree is called a full m-ary tree if every internal vertex has exactly m children.
- For example: Binary tree - every vertex has at most 2 children.



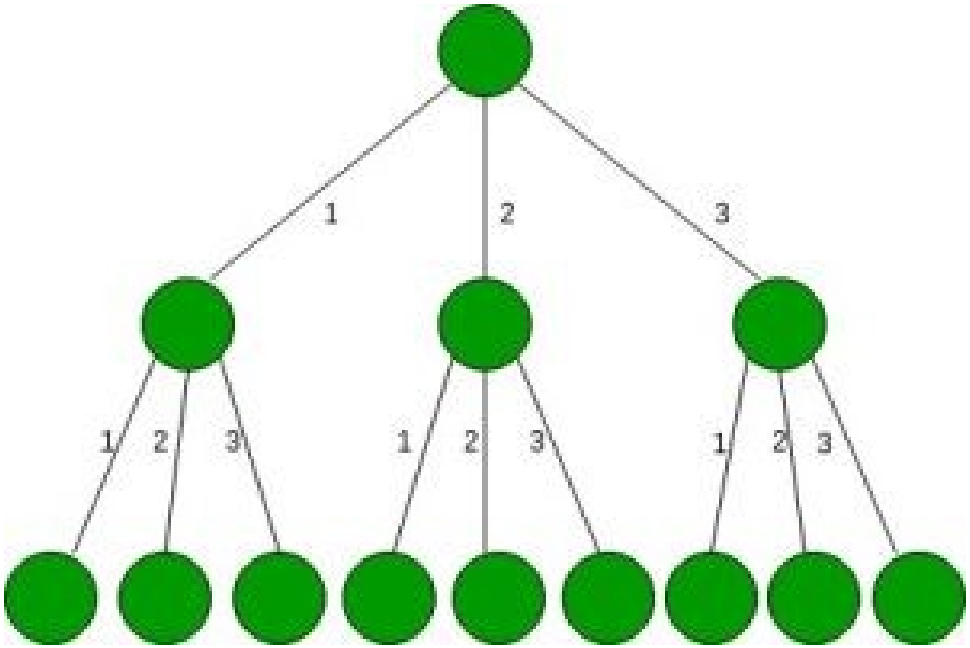
3 m-ary trees. What are the values of m for each case?

Some properties of trees

- A tree with n vertices has $n-1$ edges.
- A full m-ary tree with i internal vertices contains $n = mi + 1$ vertices.
- There are at most m^h leaves in an m-ary tree of height h.

  (height: number of levels excluded root)

Example: a full 3-ary tree



Each parent has exactly 3 children : full ternary tree

4 internal vertices, so total vertices = (3)(4)+1 = 13

13 vertices: 13-1 = 12 edges

2 levels → Not more than 3^2 = 9 leaves